

Abhishek Mahato

Full-Stack Software Engineer · React · Node.js · TypeScript · AI Engineering

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SUMMARY

Full-stack engineer with **3+ years** shipping production-grade systems — from React & TypeScript frontends to Node.js microservices, real-time data pipelines (Kafka, WebSockets), and geospatial platforms. Led delivery of platforms serving **1M+ learners across 51 countries**. Published two open-source npm libraries (schema-validex, flashcss-vite) adopted by the developer community. Now expanding into **AI engineering** (LangChain, RAG, embeddings) to build intelligent, production-ready systems.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES2024+), Python, Java, C#, .NET	Frontend	React 19, Next.js 15, Redux Toolkit, Zustand, RTK Query, React Query, Tailwind CSS, ShadCN/UI
Backend	Node.js, Express, NestJS, REST, GraphQL, gRPC, WebSockets, WebRTC	Data & Infra	MongoDB, PostgreSQL, MySQL, Redis, Kafka, Docker, CI/CD (GitHub Actions, GitLab)
AI / ML	LangChain, HuggingFace, RAG pipelines, Vector DBs (Chroma, Pinecone), Embeddings	GIS / Spatial	OpenLayers, MapLibre GL, geospatial data viz, real-time map rendering
Testing	Jest, Playwright, Cypress, Selenium — unit, integration, E2E	Tooling	Git, GitHub, GitLab, Postman, Figma, Jira, Vite, Turborepo, Nx

PROFESSIONAL EXPERIENCE

Innovate Tech *Software Developer* 10/2024 – Present | Kathmandu, Nepal

- ▶ Architected and owned end-to-end delivery of **MST-V2** — a production e-learning platform serving **1M+ learners across 51 countries**, built with React 19, TypeScript, Tailwind CSS, RTK Query, and a **Turborepo mono-repo**.
- ▶ Engineered **Ebook Scribble** — a collaborative real-time annotation system using **WebRTC**, enabling simultaneous multi-device writing, highlighting, and mobile-to-TV interaction with <150ms latency.
- ▶ Designed event-driven backend with **Kafka** for real-time streaming and **Redis** caching, reducing backend response times by **~40%** under peak load.
- ▶ Reduced page load time by **45%** by migrating data-fetching layer from Redux Saga to **React Query + RTK Query**, eliminating ~3,000 lines of boilerplate.
- ▶ Built a shared UI component library across 5 product teams, cutting cross-team feature delivery time by **~35%**.
- ▶ Contributed to **KungFu Quiz** platform integration with MST — quiz engine backed by MongoDB aggregation pipelines for dynamic question filtering and analytics.
- ▶ Conducted performance benchmarking and automated E2E test coverage using **Jest and Playwright**.

Innovate Tech *Full-Stack Trainee* 07/2024 – 09/2024 | Kathmandu, Nepal

- ▶ Delivered core features for **My Second Teacher** and **HomeSchool Asia** — platforms deployed across 8+ countries serving thousands of active students.
- ▶ Migrated legacy **class components to functional architecture** (hooks + React Query), reducing bundle size by **~18%** and improving render performance.
- ▶ Implemented MongoDB aggregation pipelines for dynamic filtering, reporting, and analytics features.

- ▶ Shipped geospatial dashboards for municipalities and NGOs using **OpenLayers** and **MapLibre GL** — high-performance real-time map rendering with complex spatial data overlays.
- ▶ Optimised state management with **React Query** + **Redux Toolkit**, cutting re-render count by ~60% in map-heavy views.
- ▶ Built reusable UI components with **ShadCN/UI**, improving design consistency and onboarding speed.

- ▶ Contributed to 10+ React components across geospatial survey and digital-addressing applications.
- ▶ Implemented lazy loading and code splitting, reducing initial bundle size by ~22%.
- ▶ Practised Git-based collaborative workflows with pull requests and structured code review.

OPEN SOURCE — PUBLISHED NPM PACKAGES

schema-validex  *TypeScript · Schema Validation*

npmjs.com/package/schema-validex · github.com/abhishek-mahato0/validex

- ▶ TypeScript-first schema validation library inspired by Zod — custom & nested schemas, per-field error reporting, transform pipelines, refinement functions, and zero-dependency runtime. Published to npm.
- ▶ Authored a technical deep-dive: "[I Built My Own Zod-Like Validation Library](#)" (8-min, Medium).

flashcss-vite  *TypeScript · Vite Plugin · CSS Engine*

npmjs.com/package/flashcss-vite · github.com/abhishek-mahato0/flashcss

- ▶ Utility-first CSS engine as a Vite plugin — on-demand class generation (UnoCSS/TailwindCSS-style), full HMR support, responsive utilities, and theme tokens with a minimal production bundle.
- ▶ Published a full engineering walkthrough: "[How I Created My Own UnoCSS-style Engine as a Vite Plugin](#)" (12-min, Medium).

KEY PROJECTS

MySecond Teacher (MST) *React 19, Node.js, MongoDB, PostgreSQL, WebSocket, WebRTC, Redis, Kafka*

- ▶ Multi-award-winning e-learning platform serving **1M+ users across 51 countries**; led full-stack delivery of the V2 rewrite — WCAG-compliant, offline-first for low-bandwidth regions, real-time collaborative annotation.

HomeSchool Asia *React, NestJS, PostgreSQL, Kafka, Redis, Docker* homeschool.asia

- ▶ Scalable homeschooling platform with real-time virtual classrooms (Kafka + WebSockets), diagnostic assessments, progress tracking, and subscription payment integration serving thousands of Cambridge IGCSE students.

License Appointment System *Next.js, MongoDB, NextAuth, ShadCN/UI, Zod, Vercel CI/CD*

github.com/abhishek-mahato0/license-appointment-system

- ▶ Role-based scheduling platform with real-time slot management, admin dashboards, audit logs, and JWT-secured APIs — deployed on Vercel with automated CI/CD pipelines.

EDUCATION

Islington College *BSc (Hons) Computing — Affiliated with London Metropolitan University* 09/2021 – 09/2024 | Kathmandu, Nepal

Global College *SLC (+2) Science* 04/2019 – 04/2021 | Kathmandu, Nepal

TECHNICAL WRITING — MEDIUM

[I Built My Own Zod-Like Validation Library in JavaScript — Here's What I Learned](#)

8-min read

[How I Created My Own UnoCSS-style Engine as a Vite Plugin](#)

12-min read

[Setting Aliases in Your Terminal](#)

5-min read